PROJECT REPORT

ON

AUTONOMOUS EV WIRELESS CHARGING AND MAGNETIC ALIGNMENT

A project report submitted in partial fulfilment of the requirements
for the project work in

DEPARTMENT OF ELECTRONICS AND TELECOMMUNICATON ENGINEERING

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ABSTRACT

Wireless Power Transfer (WPT) is a promising method for charging electric vehicles without a physical cable connection. However, the efficiency of inductive wireless charging depends strongly on the relative position of the transmitter and receiver coils. Misalignment between the coils can reduce magnetic coupling and therefore reduce the power transferred to the vehicle.

This project presents an embedded prototype for magnetic alignment assistance in an EV wireless-charging setup. An MSPM0H3216 LaunchPad acts as the controller. A TMAG5273 three-axis Hall-effect magnetic sensor measures the magnetic field along the X, Y and Z axes. The controller reads the sensor through I²C, filters the measurements, compares the X-axis reading with a calibrated target value and adjusts an SG90 servo motor using PWM. An INA219 power-monitoring module measures bus voltage and power, while an SSD1306 128 × 64 OLED displays sensor values, servo angle, alignment state, bus voltage and power information.

The firmware uses a calibration routine, a moving-average filter, alignment thresholds with hysteresis, bounded servo steps and a periodic display update. The prototype demonstrates the sensing, embedded control and monitoring functions required for an automatic alignment concept. The supplied material does not establish a complete high-power EV charging stage or measured charging-efficiency improvement; these are identified as future validation work.

Keywords: Electric Vehicle, Wireless Power Transfer, Magnetic Alignment, TMAG5273, MSPM0H3216, INA219, SG90 Servo, I²C, PWM, OLED.

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1. INTRODUCTION

1.1 Overview

Electric vehicles are increasingly adopting alternative charging methods to improve user convenience. Wireless Power Transfer allows energy to be transferred between a ground-side transmitter coil and a vehicle-side receiver coil through a magnetic field. Unlike plug-in charging, the vehicle does not need a direct electrical cable connection to the charging pad.

A key requirement for effective wireless charging is correct positioning of the two coils. When the coils are laterally or angularly displaced, the magnetic coupling may decrease. A practical charging station can therefore benefit from a mechanism that detects relative position and provides alignment assistance.

1.2 Project Concept

The proposed system uses a magnetic pilot beacon near the charging pad and a TMAG5273 three-axis Hall-effect sensor near the moving/receiver side. The sensor readings are processed by the MSPM0H3216 microcontroller. The firmware uses a calibrated X-axis magnetic reading as its target and commands an SG90 servo to make small mechanical corrections. The INA219 and OLED provide electrical monitoring and local status display.

1.3 Scope

The implementation represented by the supplied code is an embedded magnetic alignment and monitoring prototype. It demonstrates magnetic sensing, calibration, filtering, single-axis servo correction and display of selected electrical measurements. It should not be interpreted as a complete EV charger: the presentation and code do not document a high-power inverter, resonant compensation network, transmitter/receiver coil design, battery charging controller or full vehicle integration.

2. PROBLEM DEFINITION AND OBJECTIVES

2.1 Problem Statement

Wireless Power Transfer efficiency in electric vehicles decreases when the transmitter and receiver coils are misaligned. The project aims to develop a closed-loop system that detects the magnetic field using a 3D magnetic sensor, automatically adjusts alignment using a servo motor, and monitors voltage, current and power to evaluate charging performance.

2.2 Objectives

- Detect the magnetic field in three axes using the TMAG5273 Hall-effect sensor.
- Interface the magnetic sensor, INA219 and OLED with the MSPM0H3216 controller through I²C.
- Calibrate a reference magnetic position by averaging sensor readings at the desired mechanical alignment.
- Reduce sensor fluctuations using a 16-sample moving-average filter in the supplied firmware.
- Calculate the X-axis alignment error and make bounded servo corrections using PWM.
- Indicate aligned/aligning status and display magnetic readings, servo angle, bus voltage and power on the OLED.
- Establish a reference power reading when the system first enters the aligned state, where valid power data is available.

2.3 Expected Outcome

The expected outcome is a working prototype in which magnetic-field feedback is used to guide a servo-driven alignment mechanism toward a calibrated target position. The display provides live information to help observe the alignment process and the monitored electrical quantities.

3. BACKGROUND AND WORKING PRINCIPLE

3.1 Wireless Power Transfer

Wireless Power Transfer transfers energy through electromagnetic coupling between coils. In an EV charging arrangement, a transmitter coil is installed in or near the ground charging pad and a receiver coil is mounted on the vehicle. The amount of coupled energy depends on coil geometry, spacing, orientation, operating frequency and relative position. Coil misalignment can reduce coupling and lower the transferred power.

3.2 Hall-Effect Magnetic Sensing

The TMAG5273 is a three-dimensional Hall-effect magnetic sensor. It measures magnetic-field components along the X, Y and Z axes and communicates digital measurements to the microcontroller using I²C. A magnetic pilot beacon is placed near the charging pad. Changes in the sensor readings provide information about the relative magnetic position.

The project presentation notes that converting magnetic-field readings into actual coil displacement requires calibration and a known magnetic-field arrangement. Accordingly, this prototype uses a calibrated target reading rather than claiming a universal conversion from sensor counts to millimetres.

3.3 Closed-Loop Alignment

During calibration, the servo is moved to its centre position and the system records 64 X-axis sensor readings. Their average becomes targetX. During operation, the controller filters the live X-axis measurement and calculates error = filteredX − targetX. If the error is outside the alignment tolerance, the firmware calculates a small servo step and updates the PWM command. If the error is within the allowed range, the system reports an aligned condition.

3.4 Electrical Monitoring

The INA219 module is used to monitor the DC bus voltage and power. The firmware reads the INA219 bus-voltage and power registers and converts the raw readings into millivolts and milliwatts. When alignment is first achieved and a non-zero power reading is available, the firmware stores it as a reference. The OLED can show the measured power and a relative power percentage against this reference.

4. SYSTEM REQUIREMENTS AND COMPONENTS

Component	Role in the prototype	Interface / Details
MSPM0H3216 LaunchPad	Main embedded controller; runs sensor acquisition, filtering, alignment logic and display functions.	I²C controller and timer PWM; TI Driver Library configuration.
TMAG5273 3D Hall sensor	Measures magnetic-field components X, Y and Z for alignment feedback.	I²C address 0x35; sensor supply range in presentation: 1.7–3.6 V.
SG90 servo motor	Provides mechanical movement for	PWM output on PB8 /

	alignment correction.	PWM_0_INST / CC0; nominal command range limited in firmware.
INA219 power monitor	Measures bus voltage and power for charging-performance observation.	I²C address 0x40.
SSD1306 OLED display	Displays alignment state, X/Y/Z readings, servo angle, power, bus voltage and target X.	128 × 64 display; I²C address 0x3C.
Magnetic pilot beacon	Provides a magnetic reference field near the charging pad.	Physical arrangement and calibration are required.
Wiring and power supply	Provides electrical connections and power to the controller and modules.	Use module-specific voltage and current requirements.

4.1 Software and Development Environment

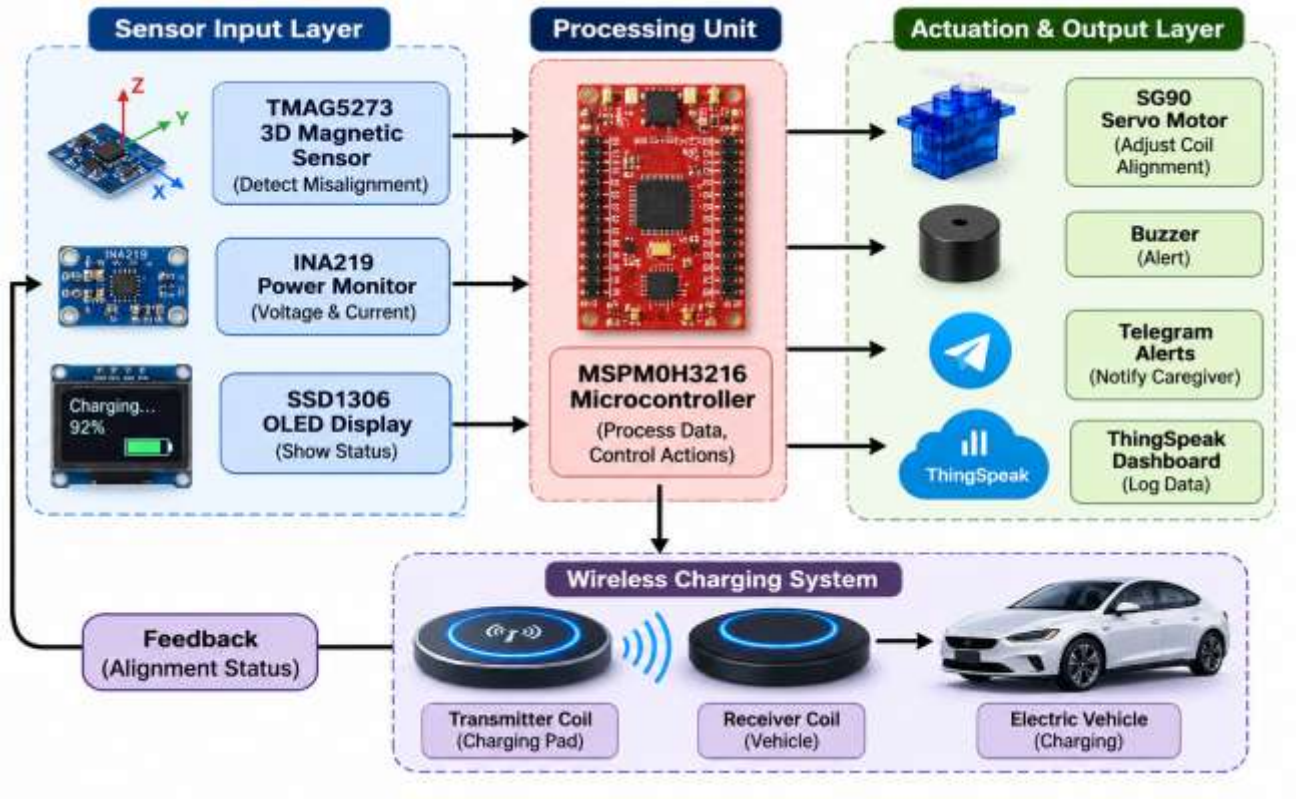
- Programming language: Embedded C.
- Header and device support: ti_msp_dl_config.h and TI Driver Library (DL) functions.
- Communication: I²C for TMAG5273, INA219 and SSD1306 OLED.
- Actuator control: Timer-generated PWM for the SG90 servo.
- Firmware structure: peripheral initialization, calibration, repeated sensor acquisition, filtering, control, monitoring and display.

4.2 Main Firmware Addresses

Device	I²C address
SSD1306 OLED	0x3C
TMAG5273	0x35
INA219	0x40

5. SYSTEM ARCHITECTURE AND CONNECTIONS

5.1 Functional Architecture



5.2 Communication and Signal Flow

The TMAG5273, INA219 and SSD1306 share the I²C bus in the firmware, each using its configured address. The controller obtains the magnetic readings from the TMAG5273, obtains bus voltage and power from the INA219, and sends display commands/data to the OLED. The SG90 servo is driven separately using a timer PWM output.

5.3 Pin Connections

Signal / Device	Controller connection stated in code	Notes
I²C SDA	PB3	Shared I²C data line for the I²C peripherals.
I²C SCL	PB2	Shared I²C clock line for the I²C peripherals.
Servo signal	PB8 / PWM_0_INST / CC0	PWM signal for SG90 control.
TMAG5273	I²C bus; address 0x35	Power and ground must follow the sensor/module specifications.
INA219	I²C bus; address 0x40	Placed in the monitored power path as required by the module design.
OLED	I²C bus; address 0x3C	SSD1306 128 × 64.

The supplied presentation contains a pin-connections overview slide, but the extracted material does not provide a complete pin-by-pin wiring table for power, ground and every module pin. Confirm those connections against the actual LaunchPad pinout and module datasheets before assembling the circuit.

6. SOFTWARE DESIGN AND ALGORITHM

6.1 Initialization

The firmware first calls SYSCFG_DL_init() to configure the LaunchPad peripherals. It starts the timer used for servo PWM, initializes the OLED, configures the TMAG5273 for continuous XYZ measurement, initializes the INA219, initializes the sensor filter, sets the servo to 90 degrees and performs target calibration.

6.2 Sensor Configuration and Acquisition

The TMAG5273 is configured for continuous measurement with X, Y and Z channels enabled. The code requests the seven-byte XYZ data sequence, reconstructs the three signed 16-bit readings and returns them to the application. The sensor configuration comments describe internal 32× averaging to reduce rapid fluctuations.

6.3 Moving-Average Filter

The firmware stores recent X, Y and Z samples in three buffers. Each new sample replaces the oldest buffer entry. The program calculates the average of the available samples (up to 16) and uses filteredX, filteredY and filteredZ for subsequent processing and display.

Moving-average equation:

$$\textbf{Filtered value} = \frac{\textbf{(sum of the latest N samples)}}{N}, \quad N \leq 16$$

6.4 Calibration

1. The servo is commanded to its centre position of 90°.
2. The OLED displays calibration instructions asking the user to keep the system still.
3. The controller collects 64 TMAG5273 readings and accumulates the X-axis values.
4. The average of the valid X readings is stored as targetX.
5. The alignment state and stored reference-power validity are reset.

6.5 Alignment Error and Servo Control

The alignment error is calculated as:

$$\textbf{error} = \textbf{filteredX} - \textbf{targetX}$$

The current firmware uses two thresholds: ALIGN_ENTER_LIMIT = 80 and ALIGN_EXIT_LIMIT = 180. This creates hysteresis. When the system is not aligned, it enters the aligned state when the error is within ±80. Once aligned, it remains aligned until the error moves beyond ±180. This prevents rapid state changes caused by small sensor fluctuations.

When alignment is not achieved, the servo-control routine is rate-limited using SERVO_CONTROL_DIVIDER. The requested step is based on error / SERVO_ERROR_DIVISOR, with a minimum step of one degree and a maximum step of three degrees. The servo angle is constrained between 10° and 170° before the PWM compare value is updated.

6.6 Servo PWM Generation

The code uses a 20,000 µs servo period. The pulse width is mapped approximately from 1,000 µs at 0° to 2,000 µs at 180°. The compare value is calculated from the period and pulse width and written to the timer capture/compare channel. The configured output is PB8 / PWM_0_INST / CC0.

6.7 INA219 Measurement

The INA219 is configured through its I²C registers. The bus-voltage reading is shifted and scaled to millivolts. The power register is multiplied by 2 to obtain milliwatts according to the code's stated power LSB. The main loop stores a non-zero power reading as the reference when the alignment state first becomes true.

6.8 OLED Display

The SSD1306 OLED is updated periodically rather than on every sensor iteration. The display routine presents the alignment state, filtered X, Y and Z readings, servo angle, power in watts, relative power percentage when a reference is valid, bus voltage in millivolts and calibrated target X.

6.9 Main Algorithm

6. Initialize the MSPM0H3216 system and start the servo PWM timer.
7. Initialize OLED, TMAG5273, INA219 and filter buffers.
8. Move the servo to 90° and calibrate targetX using 64 magnetic readings.
9. Repeatedly read raw X, Y and Z values from the TMAG5273.
10. Update the 16-sample moving-average filter.
11. Calculate alignment error from filtered X and targetX.
12. Apply hysteresis and, if required, adjust the servo by a limited step.
13. Read bus voltage and power from the INA219.
14. When aligned for the first time, save a valid non-zero power reference.
15. Refresh the OLED at the configured update interval and continue the loop.

7. IMPLEMENTATION DETAILS

7.1 Embedded Firmware Modules

Firmware function / module	Purpose
SYSCFG_DL_init()	Initializes MCU and configured peripherals.
OLED_Init(), OLED_Write(), OLED_Text()	Initializes and controls the SSD1306 OLED.
TMAG_Init(), TMAG_ReadXYZ()	Configures and reads three-axis magnetic data.
Filter_Init(), Filter_Update()	Maintains and averages recent XYZ samples.
INA219_Init(), INA219_Read16()	Configures the power monitor and reads its registers.
INA219_ReadBus_mV(), INA219_ReadPower_mW()	Converts bus-voltage and power register readings.
Servo_SetAngle()	Converts the requested angle into a PWM compare value.
Calibrate_TargetX()	Calculates the reference X-axis magnetic reading.
UpdateAlignment()	Determines alignment state and applies servo correction.
Display_System()	Shows system readings and status on the OLED.

7.2 Key Control Parameters

Parameter	Value in supplied firmware	Purpose

SERVO_PERIOD	20000 µs	20 ms servo PWM period.
SERVO_MIN_ANGLE / MAX	10° / 170°	Mechanical command limits.
SERVO_CENTER_ANGLE	90°	Initial and calibration position.
ALIGN_ENTER_LIMIT	80	Threshold to enter aligned state.
ALIGN_EXIT_LIMIT	180	Threshold to leave aligned state.
SERVO_ERROR_DIVISOR	300	Scales magnetic error into servo step.
SERVO_MIN_STEP / MAX_STEP	1° / 3°	Bounds each correction.
SERVO_CONTROL_DIVIDER	8	Slows servo correction frequency.
FILTER_SAMPLES	16	Moving-average buffer length.
OLED_UPDATE_CYCLES	8	Display refresh interval in sensor-loop cycles.

7.3 Power Reference

When the controller first detects the aligned state and the INA219 reports non-zero power, the firmware stores that reading as referencePower_mW. The OLED can then calculate relative power as (current power × 100) / reference power. This is a relative indication based on the captured reference and is not, by itself, a laboratory measurement of wireless-transfer efficiency.

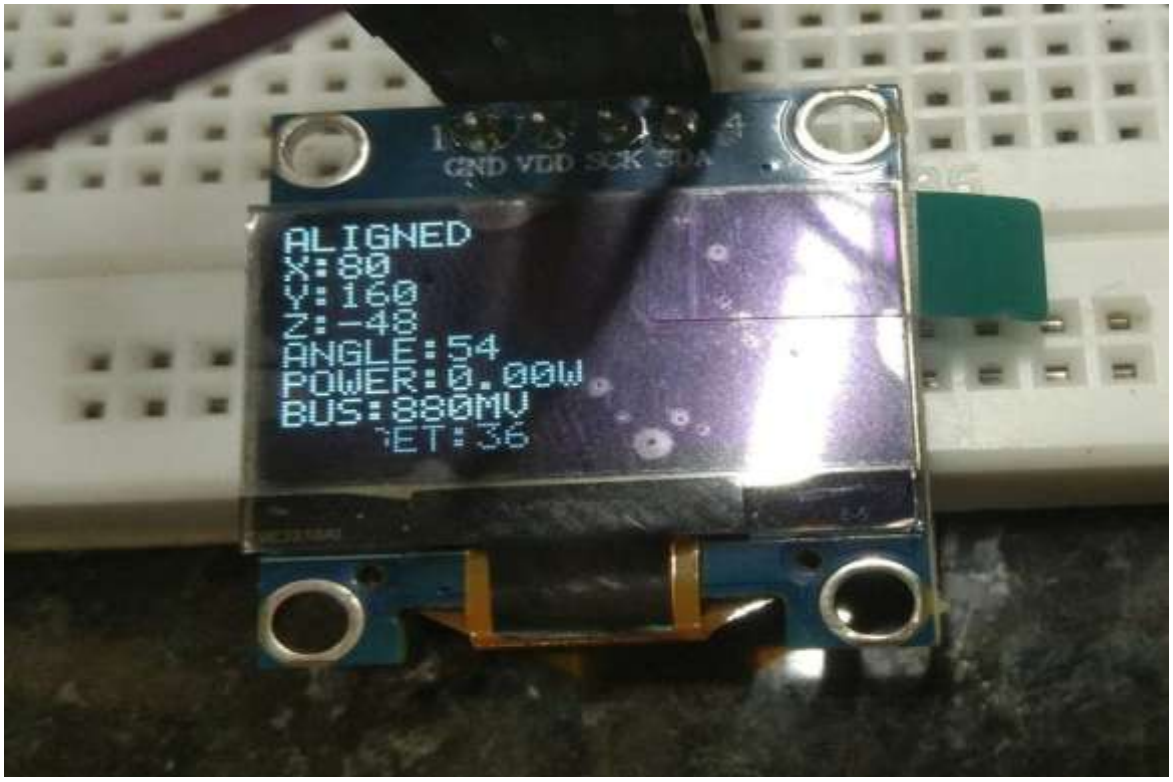
8. TESTING AND OBSERVATIONS

The supplied presentation identifies system configuration, output and working-of-project sections. The firmware also contains explicit display and control routines. However, the supplied materials do not include a complete numerical test log, measured coil displacement, measured transfer-efficiency values, or a before/after charging-power dataset. The following table is therefore a recommended validation plan rather than a claim that every test has already been completed.

Test	Procedure	Expected / observable result
I²C communication	Check that each peripheral responds at its configured address.	TMAG5273, INA219 and OLED communicate with the controller.
Magnetic sensor response	Move the sensor relative to the magnetic pilot beacon and observe XYZ readings.	Readings change with magnetic-field position and orientation.
Calibration	Place the mechanism at the desired alignment and reset the controller.	64 readings are averaged and stored as targetX.
Filter response	Compare raw readings with filtered readings while introducing small fluctuations.	Filtered values vary more smoothly than raw values.
Servo control	Introduce a controlled X-axis magnetic error.	Servo moves in bounded increments and stays within 10°–170°.
Alignment state	Bring filtered X near targetX, then move it outside the exit threshold.	OLED state changes to ALIGNED / ALIGNING using hysteresis.
Power monitor	Apply a suitable monitored DC bus and compare INA219 readings with a reference meter.	Bus voltage and power readings are available for display.
OLED display	Run the firmware through calibration and normal operation.	Display shows alignment, XYZ, servo angle, power, bus voltage and

		target X.

8.1 OUTPUT



9. ADVANTAGES, LIMITATIONS AND APPLICATIONS

9.1 Advantages

- Provides a sensor-based method for detecting magnetic position changes.
- Uses a closed-loop feedback concept rather than relying only on manual positioning.
- Filters sensor readings to reduce the effect of short-term fluctuations.
- Uses hysteresis and bounded servo movement to reduce unnecessary mechanical activity.
- Combines alignment feedback with voltage and power monitoring.
- Provides local status and measurement information through an OLED display.

9.2 Limitations

- The supplied implementation controls one servo axis using filtered X only. Y and Z are read and displayed but are not used for multi-axis correction.
- Converting magnetic-field readings into physical coil displacement requires a calibrated magnetic-field arrangement.
- The servo mechanism is a small prototype actuator; it does not establish a vehicle-scale positioning mechanism.
- The supplied project material does not document a complete high-power wireless charging transmitter/receiver stage.

- No measured charging-efficiency improvement or quantitative alignment accuracy is provided in the supplied files.
- INA219 measurements and the reference-power percentage require validation against suitable test equipment and the actual power circuit.

9.3 Applications

- Electric-vehicle wireless charging pads.
- Automatic parking and positioning for wireless charging.
- Electric wheelchairs and mobility vehicles.
- Smart parking systems.
- Robotic wireless charging.
- Fleet and public EV charging infrastructure.

10. FUTURE SCOPE

- Extend the control mechanism to two-dimensional or three-dimensional alignment using X, Y and Z sensor information and a suitable multi-axis actuator arrangement.
- Develop a calibrated mapping between magnetic-field readings and physical coil displacement in millimetres.
- Integrate a properly designed low-power wireless power-transfer demonstration stage and evaluate coupling and efficiency under controlled conditions.
- Add closed-loop optimization using measured power as an additional alignment objective, while maintaining safe operating limits.
- Replace the small servo arrangement with a mechanically suitable positioning platform for larger receiver coils.
- Add data logging, wireless telemetry and a user interface for long-term performance analysis.
- Perform repeatability, accuracy, response-time, thermal and electrical-safety tests before considering vehicle-scale deployment.

11. CONCLUSION

The project develops an embedded prototype for autonomous magnetic alignment assistance in an electric-vehicle wireless-charging concept. The MSPM0H3216 controller interfaces with a TMAG5273 three-axis Hall-effect sensor, an INA219 power-monitoring module and an SSD1306 OLED over I²C. An SG90 servo is controlled through PWM to make small corrections based on the difference between the filtered X-axis magnetic reading and a calibrated target.

The firmware includes sensor configuration, 64-sample target calibration, a 16-sample moving-average filter, hysteresis-based alignment detection, limited servo steps, power-reference capture and periodic OLED updates. These features demonstrate the core embedded sensing and control approach for magnetic alignment. Further work is required to validate physical positioning accuracy and to integrate and test a complete wireless power-transfer stage before claiming measured EV charging-efficiency improvements.

REFERENCES

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7. SG90 micro-servo technical specifications / manufacturer documentation.